

Performance (GFLOP / s)

Operational Intensity (FLOPs / Byte)

10^3

10^2

10^0

10^1

10^2

10^3

LLM Decode

GCN (Graph)

NCF (RecSys)

Autoencoder

MobileNetV2

PatchTST

DS-CNN (KWS)

DistilBERT

ResNet8

LLM Prefill

EDM Diffusion

MiniLM-L6 Qwen2.5-Co Qwen3 AST

- Hardware Ceiling (150 GB/s, 2.5 TFLOP/s)
- - - Roofline Knee (16.7 FLOPs/B)
- Memory-Bound
- Compute-Bound